

FLASHIN DELIVERY

LAST-MILE ROUTE PLANNING SYSTEM

Project report submitted in fulfilment of the requirements for the Degree

of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

By

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Registration Number: _____

Subject

INT219 - FRONTEND WEB DEVELOPMENT

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab (India)

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DECLARATION

I hereby declare that the project work entitled "**FlashIn Delivery - Last-Mile Route Planning System**" is an authentic record of my own work carried out as a requirement of the project submission for the award of B. Tech degree in Computer Science and Engineering from Lovely Professional University, Phagwara. The information furnished in this project report is based on my own study, implementation, testing, and project development work.

This project has been developed as a web-based delivery management system that supports delivery center management, driver management, customer order handling, route creation, and operational tracking for last-mile delivery.

Name of Student: Shaik Nazeer

Registration Number: _____

Date: 16 May 2026

Signature of Student

Signature of Faculty

CERTIFICATE

This is to certify that the declaration statement made by the student is correct to the best of my knowledge and belief. The student has completed the project titled "**FlashIn Delivery - Last-Mile Route Planning System**" under guidance and supervision. The present work is the result of the student's original investigation, effort, design, and study.

No part of this work has been submitted for any other degree at any university. The project is fit for submission and partial fulfilment of the conditions for the award of B. Tech degree in Computer Science and Engineering from Lovely Professional University, Phagwara.

Signature and Name of the Staff: _____

Designation: _____

School of Computer Science and Engineering,
Lovely Professional University,
Phagwara, Punjab.

Date: _____

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ABSTRACT

FlashIn Delivery is a web-based last-mile route planning and delivery management system. The main purpose of the project is to help a local delivery organization manage the movement of customer orders from stores or warehouses to final customer addresses. The system provides a centralized interface for managing delivery centers, delivery partners, customer orders, and delivery routes.

The project was developed using Laravel, Blade, Bootstrap, PHP, and SQLite. It follows a simple Model-View-Controller structure. The dashboard provides operational metrics such as total orders, delivered orders, active drivers, route counts, and recent delivery activity. CRUD functionality is provided for all major entities so that the user can create, view, update, and delete operational records.

The application is useful for college-level demonstration of full-stack web development, database design, route management, form validation, and responsive user interface design. It also demonstrates how delivery operations can be digitized to reduce manual work and improve visibility.

Keywords: Last-mile delivery, route planning, Laravel, CRUD, delivery management, dashboard, logistics.

1. INTRODUCTION

1.1 OVERVIEW OF THE WEBSITE

FlashIn Delivery is designed as an administrative web application for managing local delivery operations. The website allows the operator to maintain records of fulfillment centers, drivers, orders, and routes. It is especially suitable for grocery, essentials, and local retail delivery businesses where orders must be assigned and tracked quickly.

1.2 NEED AND IMPORTANCE OF LAST-MILE DELIVERY

Last-mile delivery is the final stage in the delivery process, where goods are transported from a hub or store to the customer. This stage is important because it directly affects customer satisfaction, delivery time, and operational cost. Manual tracking can lead to delays, duplicate work, poor assignment decisions, and limited visibility. A digital system helps the organization keep data organized and improves decision-making.

1.3 OBJECTIVES OF THE WEBSITE

- To provide a simple dashboard for monitoring delivery operations.
- To manage delivery centers, warehouses, and local stores.
- To create and maintain driver details with availability status.
- To create customer orders and assign them to drivers and routes.
- To create route records with start location, end location, distance, and estimated time.
- To provide a professional and responsive interface for project demonstration.

2. PROJECT MODULES AND FEATURES

2.1 DASHBOARD MODULE

The dashboard is the first screen of the website. It displays important summary values including total orders, delivered orders, active drivers, total routes, pending orders, assigned orders, delivery centers, recent orders, and recent routes. This gives the user a quick operational view.

2.2 MAIN MODULES

Module	Description
Delivery Centers	Stores, warehouses, and grocery hubs can be added and managed with location, contact information, type, status, and coordinates.
Drivers	Delivery partners can be stored with phone number, vehicle type, license number, assigned center, and working status.
Orders	Customer orders include order number, customer details, delivery address, items, status, assigned driver, store, and route.
Routes	Routes store route name, driver, start and end locations, coordinates, estimated distance, estimated time, and current status.

2.3 KEY FEATURES

- Responsive navigation with active page highlighting.
- Professional Bootstrap-based user interface.
- Form validation for required fields and database relationships.
- Pagination for large lists of orders, routes, drivers, and centers.
- Status badges for quick identification of delivery progress.
- SQLite database support for easy local setup.

3. LITERATURE SURVEY

3.1 LAST-MILE DELIVERY MANAGEMENT

Last-mile delivery systems focus on organizing orders, delivery partners, and dispatch routes. Existing logistics platforms show that centralized tracking reduces operational confusion and improves customer service. Delivery systems generally require a clear relationship between hubs, drivers, orders, and routes.

3.2 ROUTE PLANNING AND OPERATIONAL TRACKING

Route planning is important because delivery time and distance affect cost. Modern systems often combine route information, driver assignment, and order status. FlashIn implements the foundational version of this workflow by storing route distance, estimated time, start and end coordinates, and route status.

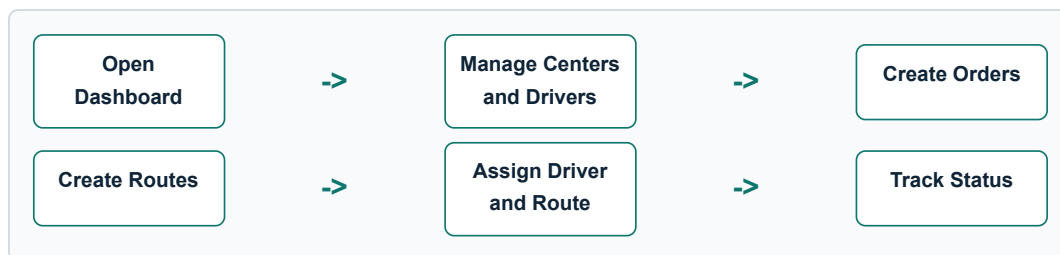
3.3 WEB-BASED CRUD SYSTEMS

CRUD applications are widely used in business software because they allow users to create, read, update, and delete records. Laravel is useful for such systems because it provides routing, controllers, validation, Blade templates, database migrations, and Eloquent ORM. These features helped in developing FlashIn as a structured and maintainable project.

4. FLOW OF THE WORK

4.1 USER JOURNEY ON THE WEBSITE

The user begins from the dashboard, where operational statistics are shown. From the navigation bar, the user can open Centers, Drivers, Orders, or Routes. The user first creates delivery centers, then adds drivers assigned to those centers. After that, customer orders can be created and associated with available centers, drivers, and routes.



4.2 NAVIGATION AND WORKFLOW

The website contains a clear top navigation bar with Dashboard, Centers, Drivers, Orders, and Routes. Every module contains listing, creation, editing, detail view, and deletion actions. The forms use Laravel validation to prevent invalid entries, and success messages are shown after important operations.

5. COMPARATIVE ANALYSIS OF DELIVERY APPLICATIONS

5.1 REVIEW OF EXISTING DELIVERY PLATFORMS

Popular delivery platforms such as Blinkit, Zepto, Swiggy Instamart, and Dunzo provide fast local delivery experiences. They include advanced features such as live tracking, automated dispatching, inventory management, payment systems, and customer mobile applications. FlashIn is a smaller academic system focused on the core administrative workflow.

5.2 FEATURE COMPARISON

Feature	Existing Apps	FlashIn Project
Order Management	Advanced customer-facing order lifecycle.	Admin can create, edit, assign, and update orders.
Driver Management	Driver apps with live status and GPS.	Admin can maintain driver records and status.
Route Planning	Automated routing and live tracking.	Manual route creation with distance and estimated time.
Dashboard	Advanced analytics and monitoring.	Simple operational metrics and recent activity.
Technology Scope	Large-scale cloud and mobile infrastructure.	Laravel web application suitable for academic demonstration.

5.3 STRENGTHS AND LIMITATIONS

The major strength of FlashIn is its simplicity. It is easy to understand, run, and demonstrate. Its limitation is that it does not yet include real-time GPS tracking, automated route optimization, customer login, or payment integration. These can be added in future versions.

6. SOFTWARE AND TOOLS UTILIZED

6.1 OVERVIEW OF SOFTWARE USED

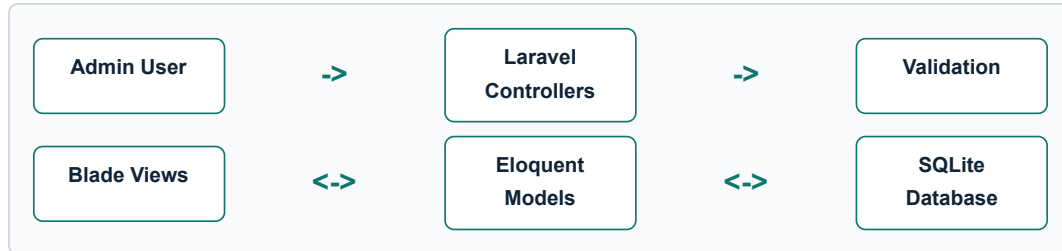
Software / Tool	Use in Project
Laravel	Backend framework used for routing, controllers, validation, models, migrations, and Blade views.
PHP	Server-side programming language for application logic.
Blade	Laravel templating engine used to build reusable web pages.
Bootstrap	Responsive layout and UI components.
Font Awesome	Icons used in navigation, dashboard, and action buttons.
SQLite	Local database used for storing project records.
Git and GitHub	Version control and remote repository hosting.
Composer	PHP dependency manager.
Vite / NPM	Frontend asset tooling available in the Laravel setup.

6.2 DATABASE DESIGN

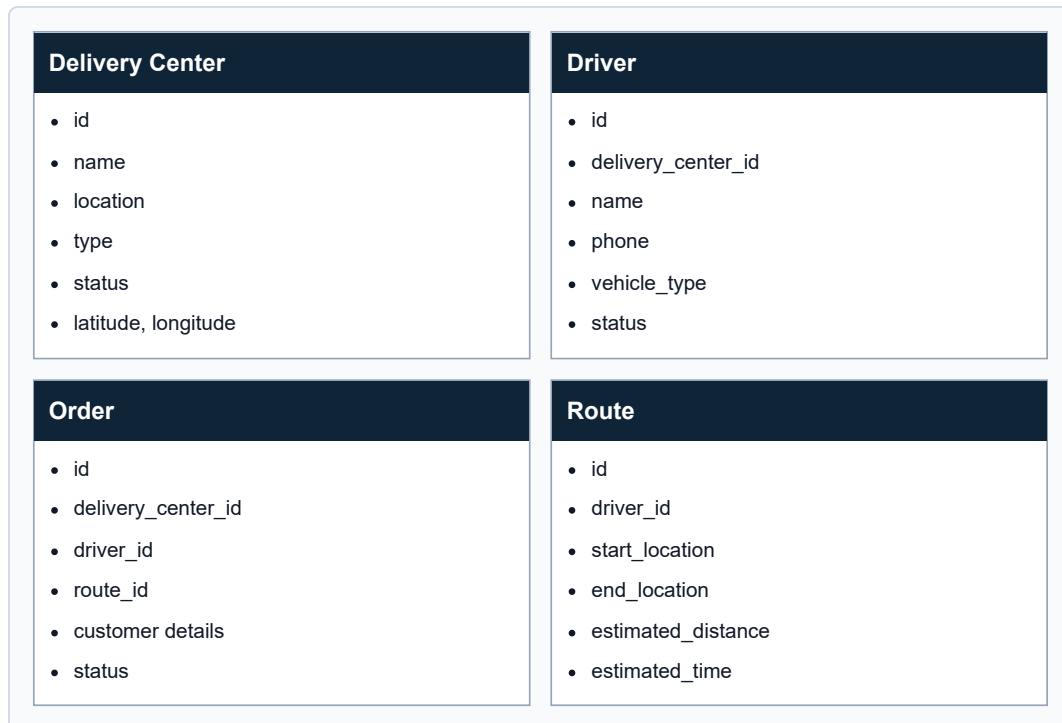
The database contains four main project tables: delivery_centers, drivers, orders, and routes. These tables are connected through foreign keys, which maintain relationships between stores, drivers, route assignments, and customer orders.

6.3 DIAGRAMS FOR WEBSITE DESIGN

DATA FLOW DIAGRAM



ENTITY RELATIONSHIP OVERVIEW



Relationship summary: One delivery center can have many drivers and many orders. One driver can be assigned to many routes and orders. One route can be connected to multiple orders through route assignment.

7. CONCLUSIONS AND FUTURE DIRECTIONS

7.1 SUMMARY OF FINDINGS

The FlashIn Delivery project successfully demonstrates a working last-mile delivery management system. It provides clear modules for centers, drivers, orders, and routes. The dashboard gives useful summary information, and the professional interface makes the website suitable for academic presentation.

7.2 RECOMMENDATIONS FOR ENHANCEMENTS

- Add user authentication and role-based access for administrators and dispatchers.
- Add automated route optimization using distance matrix APIs.
- Add live map tracking for drivers and orders.
- Add customer-side order tracking with order status updates.
- Add delivery reports, charts, and exportable analytics.
- Add payment status and invoice generation.

7.3 FUTURE SCOPE OF THE WEBSITE

In the future, FlashIn can be expanded into a complete logistics platform. Mobile apps can be added for drivers and customers. Real-time location sharing, optimized multi-stop routing, notification systems, and advanced analytics can make the system more powerful and industry-ready.

8. REFERENCES

- Laravel Documentation: <https://laravel.com/docs>
- Bootstrap Documentation: <https://getbootstrap.com/docs>
- PHP Documentation: <https://www.php.net/docs.php>
- SQLite Documentation: <https://www.sqlite.org/docs.html>
- Font Awesome Icons: <https://fontawesome.com>
- GitHub Documentation: <https://docs.github.com>

9. GITHUB LINK

Repository URL:

<https://github.com/ShaikNazeer0073/Last-mile-route-planning.git>

10. PROJECT SETUP COMMANDS

Command	Purpose
composer install	Install PHP dependencies.
copy .env.example .env	Create environment configuration file.
php artisan key:generate	Generate Laravel application key.
php artisan migrate:fresh --seed	Create tables and insert demo data.
php artisan serve	Run the project locally.